A Cost-effectiveness Analysis of Adherence Promotion Strategies to Improve Rejection Rates in Adolescent Kidney Transplant Recipients

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Rationale & Objective: Nonadherence to medical regimens increases the risk of graft loss among adolescent and young adult recipients of kidney transplants. Interventions that improve adherence may decrease rejection rates, but their perceived costs are a barrier to clinical implementation. We developed a model to assess the cost-effectiveness of an adherence promotion strategy, the Medication Adherence Promotion System (MAPS).

Study Design: Simulation-based. Data sources included published articles indexed in Medline or referenced in bibliographies of relevant English-language articles. Data on costs and outcomes were taken from a single clinical center.

Setting & Population: US adolescent patients after their first kidney transplant.

Intervention: Usual posttransplant care versus usual care plus MAPS.

Outcome: Effectiveness measured in quality-adjusted life years (QALYs) and costs measured in 2020 US dollars.

Model, Perspective, & Timeframe: Markov state transition decision model. We used a health care system perspective with a lifelong time horizon.

Results: In the base-case analysis, MAPS was more effective and less costly than usual care. MAPS cost $9,106 per patient less than usual care and resulted in a gain of 0.32 QALYs. In probabilistic sensitivity analyses, MAPS was cost saving 100% of the time. Extending results to a program level with 100 patients, any adherence promotion intervention similar in effectiveness to MAPS would cost less than $50,000/QALY if the start-up costs were <$2.5 million and annual costs <$188,000. Strategies with costs similar to MAPS that reduce the risk of rejection by as little as 3% would also have similar cost-effectiveness.

Limitations: Estimates of components and costs for MAPS were based on a single center.

Conclusions: Adherence promotion strategies with costs similar to MAPS can be cost-effective as long as they reduce rejection rates by at least 3%. This model can be applied to study the cost-effectiveness of adherence promotion strategies with varying costs and outcomes.

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Suboptimal medication adherence is a challenge across chronic medical conditions at all stages of life.[1,2] It results in worse health outcomes and decreased quality of life[3-5] while increasing health care costs in the United States by an estimated $100 to $300 billion.[6] Suboptimal adherence disproportionately affects adolescents and young adults with chronic disease for whom the characteristics of normal development (eg, increased independence, risk-taking, peer influence, and changes in brain development) adversely impact medication-taking behaviors.[7-9] This presents a profound challenge for adolescents and young adults with a kidney transplant for whom adherence to immunosuppression medication is essential to prevent allograft rejection.[10-15] Forty percent of adolescents and young adults exhibit suboptimal adherence,[16] which is associated with a 3-fold increased risk of allograft loss compared with other groups.[17] This often results in a return to dialysis, leading to a 30-year decline in life expectancy and substantially reduced quality of life.[5,18] Beyond the clinical impact of transplant rejection, the economic impact is staggering. Care for an adolescent or young adult with a failed allograft is estimated to cost between 6 and 10 times that of a patient with a functioning allograft,[19] and suboptimal adherence among kidney transplant recipients is associated with a $12,840 increase in individual 3-year medical costs.[20] Thus, interventions that are effective at improving immunosuppression adherence and reducing allograft rejection have the potential to accomplish the Institute of Healthcare Improvement triple aim of improving health and patient experience at lower cost.[21]

In recent years, 3 landmark randomized clinical trials have demonstrated the efficacy of multicomponent behavior-change interventions to improve medication adherence in solid organ transplant recipients.[22-24] Although these studies were underpowered to evaluate clinical outcomes, the authors speculated that improved adherence is likely to decrease rates of adverse outcomes (eg, acute rejection, development of de novo donor-specific antibodies, and allograft failure), justifying the cost of such interventions.[23] Additionally, there is uncertainty about how best to measure adherence in clinical practice and detect whether adherence interventions lead to improvements in meaningful clinical outcomes.

As a complement to these clinical trials, we recently reported the implementation of a Medication Adherence Promotion System (MAPS) used for all patients on immunosuppression in our kidney transplant clinic.[25,26]

PLAIN-LANGUAGE SUMMARY

Among teens with a kidney transplant, taking medication can be challenging, and poor adherence increases the risk of rejection and loss of the transplant. Strategies to help patients take medication exist, but some think these strategies are too costly or insufficiently effective. This study describes a clinical approach to improve adherence called the Medication Adherence Promotion System (MAPS) and evaluates its cost-effectiveness by using the costs and outcomes of a single center. Our findings show that this adherence intervention reduced costs and improved outcomes. Other interventions targeting improved adherence that had costs similar to MAPS can still be cost-effective if they decrease the number of rejection episodes even by a small amount.

MAPS is a practice-level intervention that focuses on identifying patients with barriers to medication adherence and partnering with them to overcome those barriers. Essential components include (1) electronic health record (EHR)-supported adherence risk screening, (2) systematic assessment of barriers to taking medication, (3) patient-centered shared decision-making aids to address identified barriers, (4) designation of team member responsibility to address specific barriers, (5) adherence promotion training for the clinical team, and (6) optional electronic medication monitoring.[26] MAPS was associated with a sustained decrease in the average acute rejection rate in our population of patients from 1.61 per 100 patient-months before MAPS implementation to 0.88 per 100 patient-months after MAPS.[26]

One barrier to widespread clinical implementation of adherence promotion strategies noted in the previously mentioned studies is their perceived high (or unknown) cost.[27] The objective of our study was to develop a framework for evaluating the cost-effectiveness of evidence-based adherence promotion strategies to reduce kidney transplant rejection rates. We used center-based experience with start-up and ongoing costs associated with MAPS as a case study that can be applied to other adherence promotion strategies in the published literature. Given that these interventions may not have the same costs or clinical outcomes when implemented in different settings and contexts, sensitivity analyses were performed to explore the implications of a range of costs and program effectiveness.

Methods

We developed a decision analytic model to simulate events and outcomes after a first kidney transplant for a prototypical 15-year-old with kidney failure. We investigated 2 strategies: (1) usual care and (2) care with MAPS integrated into the transplant clinic. The outcome measures included 2020 US dollars and quality-adjusted life expectancy measured in quality-adjusted life years (QALYs). Quality-adjusted life expectancy is a measure of effectiveness that incorporates both duration of survival, or life expectancy, and the quality of life during those years. It is an established metric of effectiveness in cost-effectiveness and cost-utility analyses.[28]

Review of Data

Data are provided with 95% confidence intervals and clinically plausible ranges when available (Table 1). We modeled the natural history of kidney disease after first transplantation using data from the published literature. Annual excess mortality rate after kidney transplantation is 0.02/year.[18] Medication adherence programs may improve adherence with immunosuppression regimens[23,24] and may decrease the rate of acute rejection.[26] In a study of 134 pediatric patients over 4 years, integrating MAPS was associated with a 45% decrease in biopsy-proven acute rejection.[26] From these data we calculated annual rates of acute rejection under usual care (0.19) and MAPS (0.11) and estimated that rejection rates decrease as a function of time from transplant by 13% each year for patients that have not experienced acute rejection.[26] Because a 45% decrease in biopsy-proven acute rejection may not be similar across varying contexts using MAPS or other adherence promotion strategies, we explored this parameter in deterministic sensitivity analyses. Among pediatric recipients of kidney transplants, acute rejection is predominantly T-cell mediated (TCMR), roughly 76%.[26] In older patients, particularly in settings of retransplantation where antibodies may have formed, the probability of TCMR versus antibody-mediated rejection (AMR) changes over time, with TCMR predominating during the first 9 months after transplantation and AMR predominating thereafter.[29] We estimated graft loss after TCMR to be 0.04/year[30] and after AMR to be 0.07/year.[31] In addition, there is a lower persistent level of graft loss ($\sim$0.01/year) that is not related to acute rejection and may not be impacted by adherence interventions such as MAPS.[31] Among patients with graft loss, the annual rate of second retransplantation is estimated to be 0.14/year.[32]

Description of Decision Model

We used Decision Maker software[33] to develop a 16 state Markov transition model, analyze decision trees, and perform sensitivity analyses using a lifelong time horizon. This type of decision analytic model allows us to perform a simulation of outcomes as patients move across different health states based on probabilistic events over time. We considered 2 strategies (see Fig 1) for a prototypical 15-year old pediatric kidney transplant recipient: (1) usual care and (2) care with MAPS.

Patients in both strategies enter the Markov simulation 3 months after transplantation. With each 3-month "tick" of the clock, patients move from 1 health state to another

Table 1. Data Required in the Analysis: Probabilities, Rates, Costs, and Quality of Life

	Base-Case Value or Cost	Distribution Type Used in Probabilistic Sensitivity Analysis
Mean age of KTx recipients, y	15	
Cumulative probability of 30-d mortality post-KTx	0.0085 (0.002-0.017)[48]	β
Annual excess mortality rate[a]		
Kidney failure receiving HD	0.074 (0.073-0.075)[18,34]	β
After KTx	0.02 (0.019-0.021)[18]	β
Rate of acute rejection (per year)		
After MAPS	0.1056 (0.0704-0.4096)[26]	Logit
Usual care	0.1932 (0.1288-0.5912)[26]	Logit
RR of acute rejection: MAPS vs usual care	0.55	
RR of acute rejection per year[b]	0.87	
Probability of TCMR[c]		
After initial KTx	0.76[26]	
After second KTx[29]		
6 wk-6 mo	0.88	
6 mo-12 mo	0.63	
>12 mo	0.08	
Probability of AMR[c]	Per[29] calculated as: 1 − Probability of TCMR	
Rate of graft loss after acute TCMR episode (per year)	0.0368 (0.0284-0.0368)[30]	Logit
Rate of graft loss after AMR episode (per year)	0.074 (0.0528-0.1049)[31]	Logit
Rate of graft loss due to causes other than acute rejection	0.0156 (0.0103-0.0235)[31]	Logit
Rate of re-KTx after graft loss among those receiving second KTx (per year)[d]	0.14 (0.135-0.143)[32]	Logit
Cost by disease state		
HD (annual)[e]	$101,332[18]	
KTx[e]		
One-time cost of procedure	$116,542[18]	
Subsequent years	$38,212[18]	
Cost of kidney biopsy[f]	$918.04[49]	
Cost of AMR[g]	$19,961.34	
Cost of TCMR[h]	$5,058.92	
Cost of MAPS (per patient)		
Initial program costs	$139.51	
Ongoing annual costs	$179.52	
Quality of life in CKD		
Receiving HD	0.53 (0.23-0.85)[40,41]	Logit
After KTx	0.84 (0.80-0.88)[42]	Logit

Values in parentheses are 95% CI or clinically plausible range. Abbreviations: AMR, antibody-mediated rejection; CKD, chronic kidney disease; HD, hemodialysis; KTx, kidney transplantation; MAPS, Medication Adherence Promotion System; RR, relative risk; TCMR, T-cell-mediated rejection.
[a]Excess annual mortality rates were calculated by adjusting United States Renal Data System (USRDS) all-cause mortality rates using relative survival at 10 years based on an age-matched US population sample.[50]
[b]Absolute rate of acute rejection decreases by 13% each year, thus relative risk of 0.87 is multiplied by the rate of rejection in the previous year.
[c]Probability of TCMR or AMR among those with acute rejection episode.
[d]Based on a 50.4% cumulative probability of re-KTx at 1.6 years.
[e]Year 2014 US dollars inflated to 2020 US dollars using Medical Care component of the Consumer Price Index (CPI).
[f]Cost of kidney biopsy includes renal biopsy percutaneous (CPT 50200), echo guide for biopsy (CPT 76942), and pathologic examination (CPT 88307).
[g]Cost of AMR includes 4 outpatient infusion center visits for rituximab and intravenous (IV) immunoglobulin × 1, and IV immunoglobulin × 3. Infusion center costs per visit: $886.20; rituximab 500 mg dose: $2,430.09; IV immune globulin 60 g dose: $3,496.61).
[h]Cost of TCMR includes 3-day hospitalization (DRG 700: $4,642.30), and professional costs (CPT 99223, 99232, 99238: $416.62).

depending on chance events that may occur. We calculated 3-month transition probabilities for each cycle, using the following equation: 3-month transition probability = $1 - e^{(-rt)}$, where r is the annual event rate and t is three-twelfths of a year. After transplantation, patients die or survive based upon US population mortality tables and excess mortality among kidney transplant patients.[18,34] As shown in Figure 1, we have defined separate health states for a series of "Dead" states so we could calculate at the end of the simulation the proportion of patients dying from nonexplicitly modeled causes (ie, life table–based mortality), kidney failure, and following successful

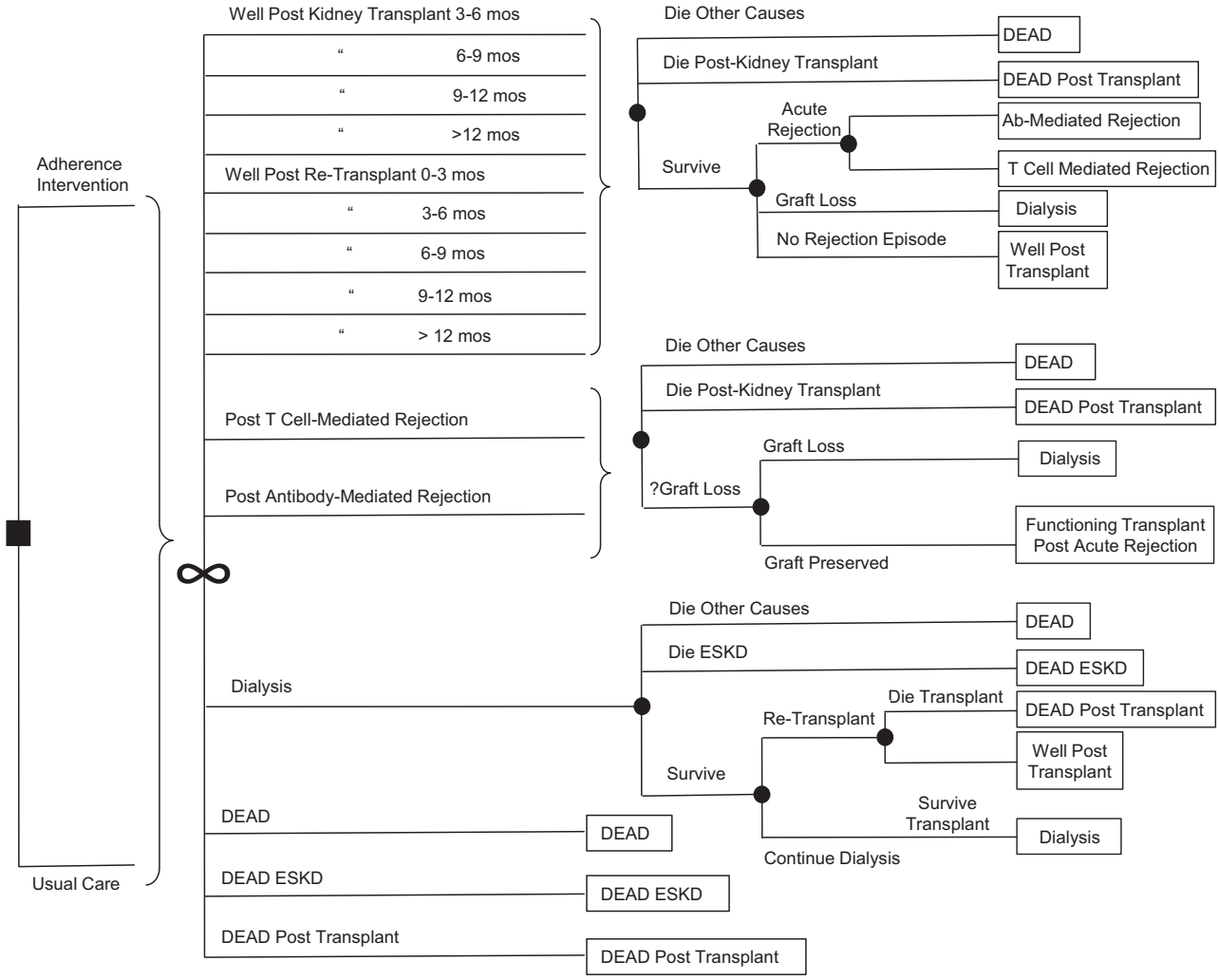


Figure 1. Decision tree model. The context of the analysis are adolescents 3 months after successful kidney transplantation for kidney failure, with an average age of 15 years. The 2 strategies shown at the square decision node include (1) usual care and (2) a medication adherence promotion system (MAPS). Round nodes represent chance events. In both strategies patients enter the Markov state transition model in the health state Well Post-Kidney Transplant 3-6 months. Patients then proceed through a series of tunnel states capturing the first 12 months after kidney transplantation. During any 3-month cycle, patients may die from nonexplicitly modeled causes, based on population life tables from the Centers for Disease Control and Prevention (CDC)[50] or face excess mortality associated with kidney transplantation. Survivors in each cycle face the risk of developing acute rejection, which may be antibody mediated or T-cell mediated. They also may have graft loss that is not related to acute rejection. Graft loss leads back to dialysis. Patients who have developed acute rejection may die from nonexplicitly modeled causes or face excess mortality risk after kidney transplantation. In addition, they face the risk of rejection-associated graft loss. Aside from mortality risk, patients receiving dialysis may receive a second kidney transplant. Abbreviations: Ab, antibody; ESKD, end-stage kidney disease.

kidney transplantation. Surviving patients face risks of acute rejection, which may be TCMR or AMR, or graft loss due to non–rejection-related causes (eg, recurrent disease, chronic allograft nephropathy, etc).

Patients proceed through a series of tunnel states during the first year after transplantation during which the relative probabilities of TCMR versus AMR change as described earlier.[29] Graft loss leads to dialysis. Acute rejection decreases allograft survival with a subsequent increased risk of graft loss and return to dialysis. After graft loss, patients receiving dialysis may receive a second kidney transplant. We model the MAPS intervention to continue after a second transplant and continue to provide protection against acute rejection.

Costs

We performed our analyses from the health care system perspective and did not include indirect costs (eg, lost wages, productivity). The costs of the intervention included those required for establishment and maintenance of MAPS, including information technology, materials purchase, and personnel time (Tables S1 and S2). The

majority of the start-up costs for this analysis are equipment costs—tablet computers and optional electronic pillbox monitoring—which overestimates these costs from a generalizability standpoint because earlier iterations of MAPS were done using low-tech strategies (ie, using paper forms in clinic to assess for adherence barriers). Ongoing costs were largely personnel costs, which factored in the amount of time spent working with patients to discuss adherence barriers and how to overcome these barriers and maintain the integration with MAPS and the EHR. Personnel costs include fringe benefits at a rate of 27.5%. Start-up costs were estimated at $139.51 per patient, with ongoing annual per-patient cost of $179.53. The costs in our simulation were spread over a population of approximately 100 patients; although these are not completely fixed for variation in population size, there are economies of scale such that a smaller program would likely cost more per patient for start-up whereas a larger program would likely cost less per patient.

The outcome costs include publicly available cost data for (1) kidney biopsies (performed as an outpatient procedure on patients with acute rejection), (2) treatment of TCMR and AMR, (3) dialysis care, and (4) post–kidney transplant care. In an effort to underestimate the benefits of adherence promotion strategies, the costs for treatment of rejection were based on the lowest intensity treatment from the published literature. Thus, the costs for treatment of TCMR were based on a hospitalization for 3 pulse doses of intravenous methylprednisolone and costs for AMR were based on a single outpatient infusion of rituximab and 4 outpatient infusions with intravenous immune globulin (IVIG).[35] This is an underestimate of costs associated with both rejection types because treatment for more severe rejections may involve prolonged hospitalizations for T-cell–depleting therapies such as antithymocyte globulin and/or treatment with plasmapheresis, proteosome inhibition, or other novel therapies.[36-38] Costs were expressed in 2020 US dollars. Future costs and effectiveness were discounted at 3% per year.[28,39]

Quality of Life

Because kidney failure has a significant impact on quality of life, the use of QALYs as a measure of effectiveness is particularly important.[5] Separate utilities were used for patients with kidney failure receiving dialysis and those post–kidney transplantation.[40-42]

Analyses

The base-case analysis examines the incremental cost-effectiveness ratio (ICER) of MAPS compared with usual care. Given the analyses are simulation-based, deterministic and probabilistic sensitivity analyses were performed to examine the impact of parameter uncertainty. Probabilistic sensitivity analyses using second-order Monte Carlo simulations were conducted using distributions for most parameter values in the decision model. Beta and logit distributions were used for probabilities and lognormal distributions for relative risks, hazard ratios, rates, and costs. Deterministic sensitivity analyses were performed by systematically varying 1 or more parameters values over clinically relevant ranges.

Results

Base-Case Analysis

In the base-case analysis, we discounted costs and effectiveness at 3% per year. As shown in Table 2, according to the model MAPS cost $9,106 per patient less than usual care and was more effective, resulting in a gain of 0.32 QALYs per patient. Without discounting, MAPS reduces costs by $3,131 and leads to a 0.8 gain in QALYs, dominating usual care. Figures S1 and S2 show movement of the patient cohort through the major health states over time from the start of the simulation. Patients in the usual care strategy develop acute rejection more quickly and have a higher peak than those in the MAPS strategy. Figure 2 shows cumulative probability of acute rejection and death-censored graft survival for both strategies. Fifty-eight percent and 38%, respectively, of patients in usual care versus MAPS experience a first episode of acute rejection by 10 years. Graft survival at 10 years is roughly 67% for patients in usual care and 70% for patients in MAPS.

Deterministic Sensitivity Analyses

One of the most important model parameters is the estimated effectiveness of MAPS as noted by the relative risk (RR) of acute rejection among those receiving MAPS versus usual care. In our base case, we used a RR of 0.55. As can be seen in Figure 3, MAPS dominates usual care by being less costly and more effective until the RR exceeds 0.88. Above a RR of 0.88 (indicating decreasing effectiveness) the ICER of the MAPS strategy increases. Above a RR of 0.95, MAPS is no longer highly cost-effective using a cutoff of $50,000/QALY.[28,43] Figure 4 examines the cost of MAPS. The top panel examines the initial start-up cost, and the bottom panel examines the ongoing yearly cost of the program. MAPS remains cost saving or highly cost-effective so long as the start-up cost attributable to a single patient is less than $25,000 and the ongoing annual program cost per patient is less than $1,880 per year. In sensitivity analyses of other model parameters within their 95% confidence intervals or clinically plausible ranges, MAPS remained cost saving.

Probabilistic Sensitivity Analysis

As shown in Figure 5, in over 10,000 iterations, MAPS was more effective than usual care 100% of the time, yielding an average gain of 0.32 ± 0.06 (SD) QALYs at an average lifetime cost savings of $9,155 ± $1,317 per patient. As

Table 2. Results of Base-Case Analysis

Strategy	Cost ($)	Effectiveness (QALYs)	Incremental Cost ($)	Incremental Effectiveness (QALYs)	Incremental Cost-Effectiveness ($/QALY)
Discounted at 3%/Year					
MAPS	804,629	13.7458			
Usual care	813,735	13.4257	9,106	−0.3200	Dominated
Not Discounted					
MAPS	1,438,515	23.9249			
Usual care	1,441,646	23.1242	3,131	−0.8007	Dominated

Abbreviations: MAPS, Medication Adherence Promotion System; QALY, quality-adjusted life years.

shown in the cost-effectiveness acceptability curve, MAPS was less costly and more effective, thus it dominated usual care 100% of the time.

Discussion

A recent Cochrane review[44] as well as a National Institutes of Health consensus conference[45] promoted the need for randomized trials of adherence promotion interventions for adolescent and young adult transplant recipients, and recommended the development and testing of multicomponent interventions. In response, 3 randomized controlled trials of multicomponent interventions have demonstrated improved medication adherence in solid organ transplant recipients.[22-24] In a clinical setting, implementation of a multicomponent adherence intervention, MAPS, was associated with a 45% reduction in acute rejection.[26] Yet the perceived or unknown cost of such interventions remains a barrier to clinical implementation.[27] In this study, MAPS was used as a case study for cost-effectiveness of adherence interventions to reduce acute rejection. Analyses demonstrated that an adolescent kidney transplant patient, 15 years of age, could gain an average 0.32 QALYs from participating in an adherence program with similar costs and effectiveness to those we estimated from MAPS. This corresponds to a decrease in acute rejection episodes from 63% to 43% over the lifetime of a patient. In addition, the average lifetime cost of MAPS is estimated to be $9,106 less per patient than usual care.

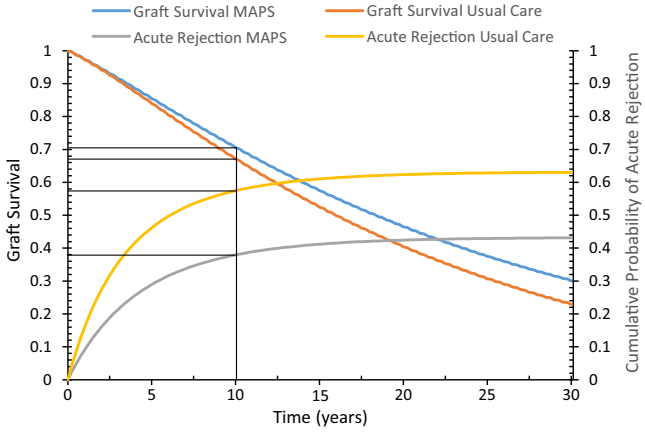


Figure 2. Graft survival and cumulative probability of acute rejection. The figure shows graft survival (left vertical axis) and the cumulative probability of acute rejection (right vertical axis) with either MAPS or usual care. At 10 years into the Markov simulation, graft survival with MAPS is just over 70% (blue line) while graft survival with usual care is 67% (orange line). The cumulative probability of acute rejection is 38% with MAPS (grey line) and almost 58% with usual care (yellow line). The *x*-axis shows time out to 30 years. Abbreviation: MAPS, Medication Adherence Promotion System.

Although the results for our particular MAPS program look promising, estimates for the cost-effectiveness of MAPS come from a small, single-center study and may vary widely in different contexts or settings. The sensitivity analyses we performed allow our model and analysis to apply to other medication adherence programs that might have different costs or outcomes (ie, different effect size in reducing rejections). Although the absolute decrease (or absolute risk reduction) in acute rejection rates of 8.8% per year (10.56% vs 19.32%) estimated from MAPS is both clinically and economically significant, sensitivity analysis shows that if start-up and ongoing costs are similar, adherence promotion strategies that lead to an absolute rejection rate decrease as small as 3% per year (ie, from 20% to 17%) would be less costly and more effective than usual care. In fact, if these strategies only decreased the rate of rejection from 20% to 18.5% per year, they would still be highly cost-effective based on the commonly accepted willingness-to-pay threshold of $50,000 per QALY.[28,43]

The estimated start-up costs of $139.51 per patient with ongoing annual cost of $179.53 per patient (Fig 4) primarily reflect personnel costs along with expenditures for equipment. Through sensitivity analysis, we find that adherence promotion strategies with effectiveness similar to those estimated for MAPS would remain *cost saving* even if start-up costs are as high as $9,300 per patient and annual ongoing costs are as high as $800 per patient. Using the data from our real-world experience, our model predicts

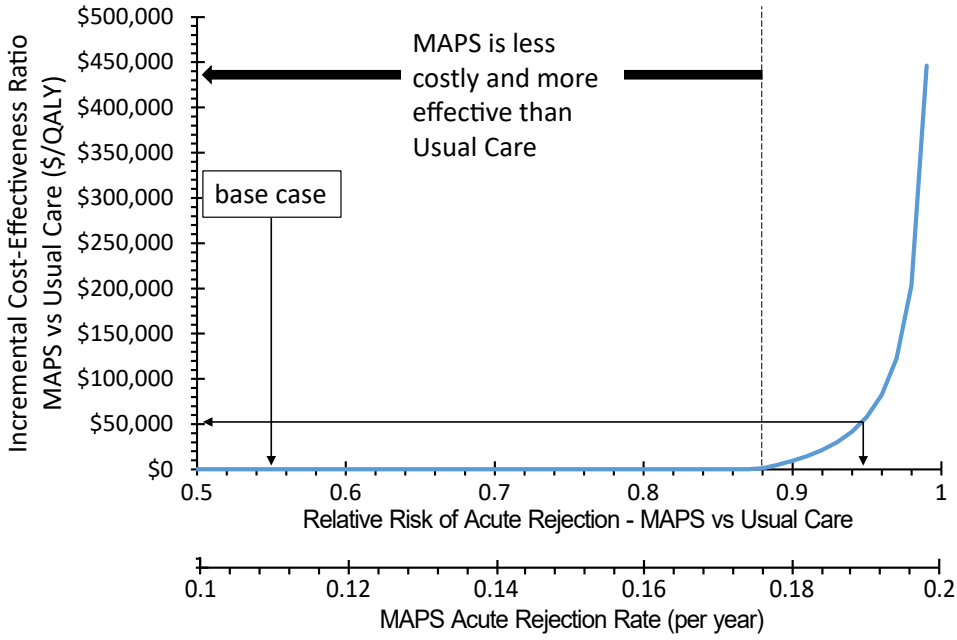


Figure 3. One-way sensitivity analysis of relative risk of acute rejection MAPS versus usual care. ICER of MAPS versus usual care is plotted as a function of the relative risk of acute rejection of MAPS versus usual care. The second *x*-axis displays the absolute annual rate of acute rejection with the MAPS strategy. At the base-case value of 0.55, corresponding to an absolute annual rejection rate of roughly 0.11, the MAPS strategy dominates usual care by being less costly and more effective. As the efficacy declines (ie, the RR increases) MAPS no longer dominates above a RR of 0.88, beyond which the ICER of the MAPS strategy increases. Above a RR of 0.95, MAPS is no longer highly cost-effective (ICER > $50K/QALY). Abbreviations: ICER, incremental cost-effectiveness ratio; MAPS, Medication Adherence Promotion System; QALY, quality-adjusted life year; RR, relative risk.

that MAPS is highly cost-effective (ie, <$50,000 per QALY) for start-up costs as high as $25,000 per patient with ongoing annual program costs less than $1,880 per patient. In other words, in a population of 100 patients, a program with similar effectiveness that costs up to $930,000 at start-up and up to $80,000 per year to maintain would actually *save* money while improving health outcomes and quality of life. Such a system in this population could have start-up costs as high as $2.5 million and ongoing annual costs of $188,000 and still be considered *highly cost-effective*. Once more, it should be noted that these costs are from the health system perspective overall and not a particular insurer, provider, or hospital. Thus, they represent as best as possible the true resource utilization costs at the aggregate health system level.[28]

The implications of this study extend well beyond the specific use case of MAPS and can be applied to any of the published interventions to improve adherence in solid organ transplant. To date, published randomized clinical trials have not been powered to examine clinical outcomes (eg, rejection or allograft function), so the cost savings or cost-effectiveness of such interventions could not be determined. A recent editorial[27] highlighting these studies acknowledged that the barriers to implementing such programs include the lack of demonstrated effectiveness in improving clinical outcomes and the perceived cost of implementation in the clinical arena. As Foster noted, "there is a belief that adherence interventions should be able to be implemented at little or no additional cost—a requirement that never applied to medical or surgical interventions."[27] Our analysis, demonstrates that even very costly interventions would still be highly cost-effective or even cost saving if there was a measurable decrease in acute rejection associated with their implementation. Thus, clinical programs designed to reduce acute rejection through improved medication adherence have the potential to promote the Institute for Healthcare Improvement's Triple Aim to (1) improve the patient experience of care by increasing quantity and quality of life, (2) improve the health of populations, and (3) reduce the per capita cost of health care.[21]

This study has several limitations. First, the cost and effectiveness estimates are from a small single-center study, which may not be generalizable given that the effectiveness of any complex intervention is related to context-specific factors (eg, quality improvement support, organizational support, other environmental factors). Yet, although the specific results may not be generalizable, the overall framework of this cost-effectiveness model should be. The benefit of sensitivity analysis is that it generalizes the specific situation to an entire range of costs and effectiveness of any intervention. Thus, as long as one is

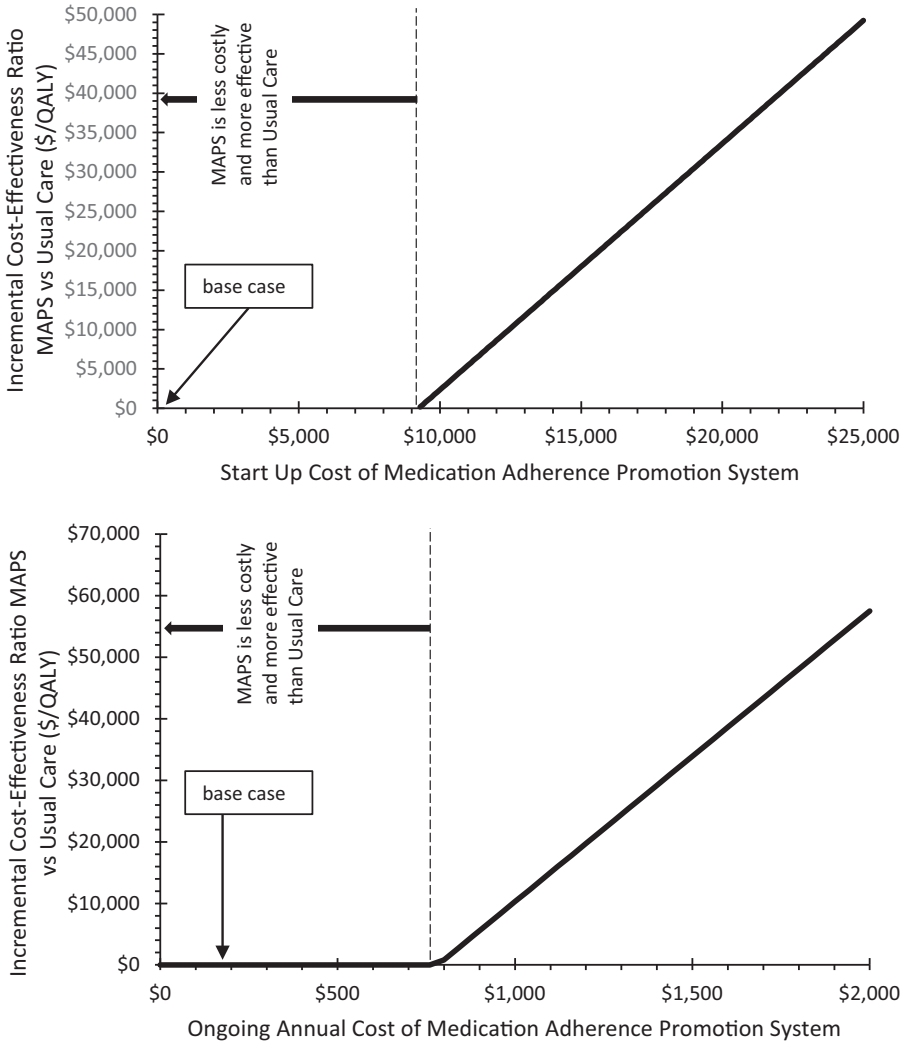


Figure 4. One-way sensitivity analysis of cost. Top panel: start-up cost of MAPS. The *x*-axis displays the initial start-up cost for the MAPS strategy. Costs include the cost of software development spread across a single facility's pediatric kidney transplant population, development of provider scripts, monitoring parameters, and discussion points, along with the purchase of monitored pillboxes. MAPS is cost saving at the base-case start-up cost of $140. MAPS remains cost saving until start-up costs exceed $9,300. MAPS remains highly cost-effective at less than $50K/QALY until start-up costs exceed $25,000. Bottom panel: ongoing annual cost of MAPS. Base-case estimate for ongoing annual costs was $180. MAPS is cost saving below an annual cost of $800 and remains highly cost-effective until annual ongoing costs exceed $1,880. Abbreviations: ICER, incremental cost-effectiveness ratio; MAPS, Medication Adherence Promotion System; QALY, quality-adjusted life year.

able to estimate specific effectiveness of an adherence intervention with start-up and ongoing costs within their organization, the results of these sensitivity analyses make this model useful in other contexts.

Second, data were used from pediatric and adolescent studies when available, but some data came from the adult literature. Specifically, robust data describing baseline rejection rates and the proportion of TCMR to AMR are lacking for pediatric transplant patients. Furthermore, it is difficult to model the effects of chronic rejection over time, although the attrition in kidney function from chronic rejection is built into the estimates of allograft loss from acute TCMR, AMR, and other causes. One way to check the assumptions of this model is to compare allograft failure rates in the model with those from Organ Procurement Transplant Network (OPTN) Scientific Registry Transplant Report (SRTR). The model reports that graft survival at 10 years is roughly 67% for patients in usual care. This finding approximates the 10-year allograft survival of 61% (deceased donor) and 73% (living donor) reported by the OPTN and provides a measure of validation for the assumptions in the model.[46]

There remains uncertainty about how best to monitor and measure medication adherence in practice and detect if

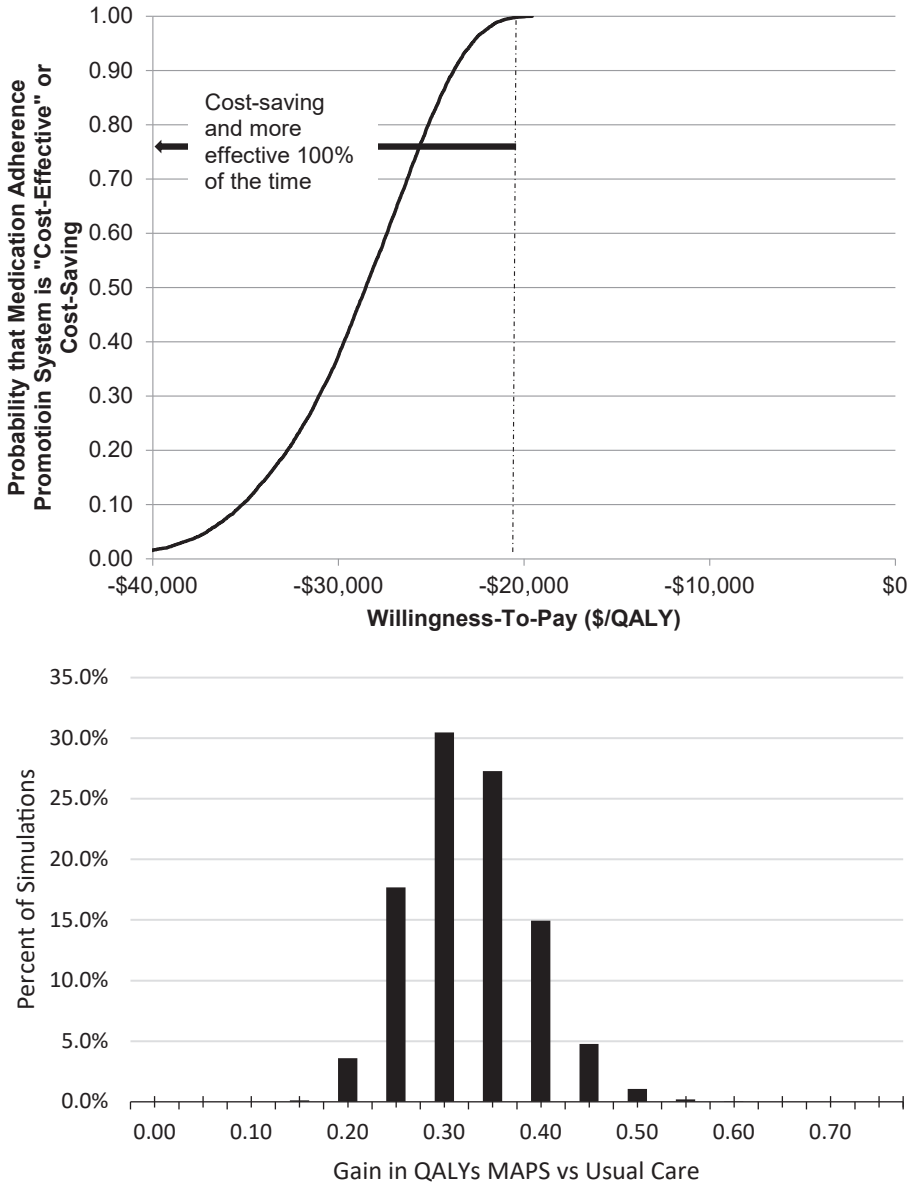


Figure 5. Probabilistic sensitivity analysis. Top panel: Cost-effectiveness acceptability curve for MAPS versus usual care. Probabilistic sensitivity analysis was done by performing 10,000 iterations of a second-order Monte Carlo simulation. MAPS is cost saving and more effective than usual care 100% of the time. Bottom panel: Distribution of gain in QALYs between MAPS and usual care over 10,000 iterations of a second-order Monte Carlo simulation. The average gain was 0.32 QALYs (SD, 0.06). Abbreviations: MAPS, Medication Adherence Promotion System; QALY, quality-adjusted life year.

interventions to increase adherence lead to improvements in rejection and graft failure rates. Although measurement of adherence is challenging and large clinical trials of adherence interventions have not been powered to assess outcomes, we drew from our experience with MAPS to populate our model with real-world cost and outcomes data. Future medication adherence research should build on studies that have already demonstrated the *efficacy* of interventions by evaluating the implementation and real-world *effectiveness* of such programs within clinical systems of care.

The present study provides a model for assessing the cost-effectiveness of adherence interventions so long as estimates of cost and effectiveness can be obtained. The findings suggest that even small improvements in rejection rates justify relatively large start-up and maintenance costs of adherence interventions. Currently, an effort is underway to study the implementation, spread, and effectiveness of MAPS to participating kidney transplant programs in the Improving Renal Outcomes Collaborative.[47] The results of this analysis can be used to inform the cost-effectiveness of these efforts and

other adherence promotion strategies in kidney transplant recipients.

Supplementary Material

Supplementary File (PDF)

Figure S1: Major health states over time for usual care cohort.

Figure S2: Major health states over time for MAPS cohort.

Table S1: Initial cost data for MAPS.

Table S2: Ongoing cost data for MAPS.

Article Information

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